

Enhancing CNN-based Image Classification Robustness and Generalization using the Push-Pull CORF Operator for Shape-Biased Learning

Master Thesis Defense by Shrushti Kaul

Supervisors: Dr. G Azzopardi and S. Esposito

Introduction

Image Classification

One of the most prominent branches of computer vision

Introduction

Image Classification

One of the most prominent branches of computer vision

ImageNet + ILSVRC

Around 14 million images spanning 22k classes

ImageNet Large Scale Visual Recognition challenge (focused on 1k classes) -

AlexNet, ResNet classifiers

Introduction

Image Classification

One of the most prominent branches of computer vision

ImageNet + ILSVRC

Around 14 million images spanning 22k classes

ImageNet Large Scale Visual Recognition challenge (focused on 1k classes) -

AlexNet, ResNet classifiers

Classifiers are “superhuman”

But are they?? Let's see what happens when we test on images outside of the ImageNet validation/test set

Introduction

Fox Squirrel



Introduction

Fox Squirrel



Fox Squirrel

Sea Lion (99%)



Introduction

Dragonfly



Introduction

Dragonfly



Dragonfly

Manhole Cover (99%)



Introduction

Performance becomes “subhuman”

When the real-time data is not similar to data used to train the models

Introduction

Performance becomes “subhuman”

When the real-time data is not similar to data used to train the models

Great ImageNet validation/test scores mislead

The validation/test set only contain simple, clear, close-up images – too easy of a benchmark

Introduction

Performance becomes “subhuman”

When the real-time data is not similar to data used to train the models

Great ImageNet validation/test scores mislead

The validation/test set only contain simple, clear, close-up images – too easy of a benchmark

Proved that CNNs under utilize shape information

Artificially overwrote ImageNet’s texture and could conclude that classifiers seem to rely on textural cues – texture-bias

Introduction – Texture Bias



(a) Texture image

81.4% **Indian elephant**
10.3% indri
8.2% black swan



(b) Content image

71.1% **tabby cat**
17.3% grey fox
3.3% Siamese cat



(c) Texture-shape cue conflict

63.9% **Indian elephant**
26.4% indri
9.6% black swan

Introduction – Texture Bias



(a) Texture image
81.4% **Indian elephant**
10.3% indri
8.2% black swan



(b) Content image
71.1% **tabby cat**
17.3% grey fox
3.3% Siamese cat



(c) Texture-shape cue conflict
63.9% **Indian elephant**
26.4% indri
9.6% black swan

Improving shape-bias can increase robustness

Artificially overwrote ImageNet's texture and could conclude that classifiers seem to rely on textural cue – texture-bias

Related Work

Diverse Data Augmentation methods to increase robustness

Related Work

Diverse Data Augmentation methods to increase robustness

Mixing-Based

AugMix: applies a random sequence of augmentations, such as rotation, scaling, or color jittering

CutMix: replaces a random region of an image with a patch from another.

Mixup: linear interpolation of 2 images (& their labels)

MoEx: During training, select 2 images and replaces their feature moments (mean and variance). The target labels of both images are combined.

Related Work

Diverse Data Augmentation methods to increase robustness

Mixing-Based

Adversary-Learning Based

AugMix: applies a random sequence of augmentations, such as rotation, scaling, or color jittering

CutMix: replaces a random region of an image with a patch from another.

Mixup: linear interpolation of 2 images (& their labels)

MoEx: During training, select 2 images and replaces their feature moments (mean and variance). The target labels of both images are combined.

Add perturbations/adversarial examples to the training set
Ex: ANT

Related Work

Diverse Data Augmentation methods to increase robustness

Mixing-Based

AugMix: applies a random sequence of augmentations, such as rotation, scaling, or color jittering
CutMix: replaces a random region of an image with a patch from another.
Mixup: linear interpolation of 2 images (& their labels)
MoEx: During training, select 2 images and replaces their feature moments (mean and variance). The target labels of both images are combined.

Adversary-Learning Based

Add perturbations/adversarial examples to the training set
Ex: ANT

Frequency-Based

Patch Gaussian: Applies Gaussian noise to specific patches.
Speckle noise: Multiplies the input image with random noise.

Related Work

Diverse Data Augmentation methods to increase robustness

Mixing-Based	Adversary-Learning Based	Frequency-Based	Other Methods
<u>AugMix</u>: applies a random sequence of augmentations, such as rotation, scaling, or color jittering <u>CutMix</u>: replaces a random region of an image with a patch from another. <u>Mixup</u>: linear interpolation of 2 images (& their labels) <u>MoEx</u>: During training, select 2 images and replaces their feature moments (mean and variance). The target labels of both images are combined.	Add perturbations/adversarial examples to the training set Ex: <u>ANT</u>	<u>Patch Gaussian</u>: Applies Gaussian noise to specific patches. <u>Speckle noise</u>: Multiplies the input image with random noise.	<u>Stylized ImageNet</u>: Applies style transfer to create stylized images. <u>AutoAugment</u>: Uses reinforcement learning for optimal augmentation policies. (search space includes various image processing operations) <u>DeepAugment</u>: Bayesian Optimization to discover optimal data augmentation hyperparameter <u>Cutout</u>: Masks out square regions of input images.

Aim

Find a method to enhance learning of shapes and not texture

Aim

Find a method to enhance learning of shapes and not texture

But how?

Aim

Find a method to enhance learning of shapes and not texture

But how?

Take motivation from well-known robust vision systems

Aim

Find a method to enhance learning of shapes and not texture

But how?

Take motivation from well-known robust vision systems

Taking inspiration on nature, the mammalian brain is adept at object recognition.

We're not fooled by many forms of corruptions

How do we see?

- The Retina: Light enters our eyes and is captured by the retina.
- The Lateral Geniculate Nucleus (LGN): Visual signals are then relayed to the LGN, a part of the thalamus that acts as a gateway for visual information.
- The Primary Visual Cortex (V1): From the LGN, signals travel to V1, where simple cells play a crucial role in detecting edges and contours.

How do we see?

- The Retina: Light enters our eyes and is captured by the retina.
- The Lateral Geniculate Nucleus (LGN): Visual signals are then relayed to the LGN, a part of the thalamus that acts as a gateway for visual information.
- The Primary Visual Cortex (V1): From the LGN, signals travel to V1, where simple cells play a crucial role in detecting edges and contours.

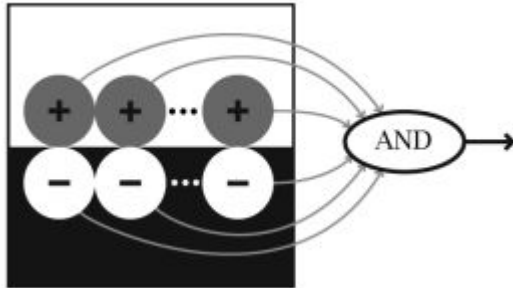
LGN cells have a fascinating structure:

- They possess center-surround receptive fields
- These fields respond differently to light in their center versus their surround
- Closest analogy: bullseye target (The center might respond positively to light, while the surround responds negatively)

Modeling the simple cells

CORF - Combination of Receptive Fields

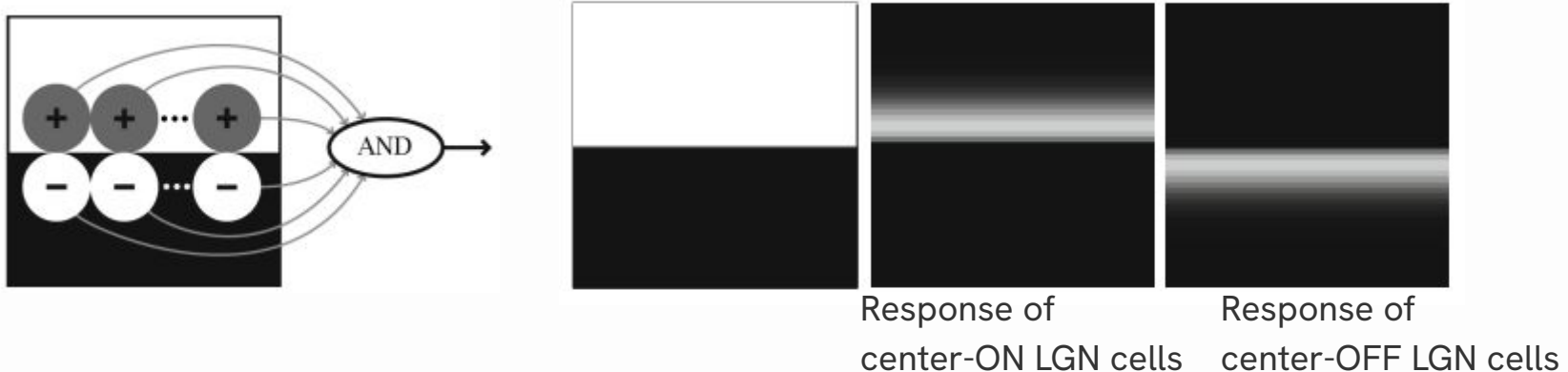
- It combines the responses of multiple model LGN cells
- These model cells are arranged in specific patterns, mimicking the structure observed in the visual cortex
- The combination is done using a weighted geometric mean (detect edges and contours based on stimulus orientation)



Modeling the simple cells

CORF - Combination of Receptive Fields

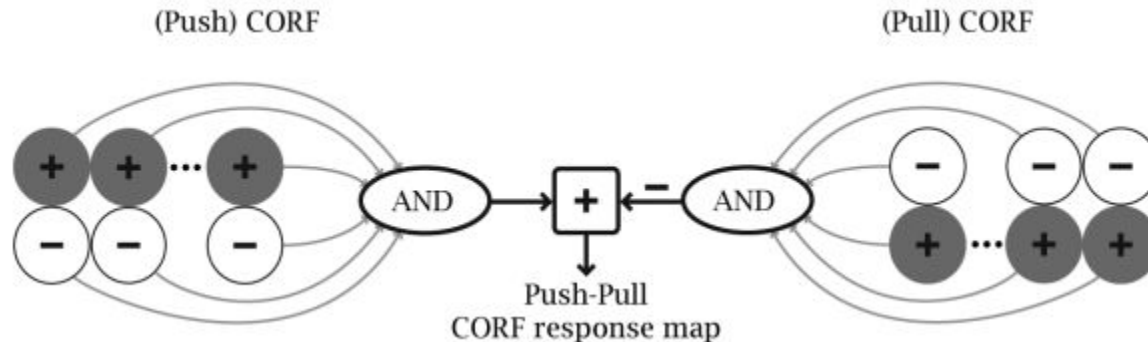
- It combines the responses of multiple model LGN cells (modelled by DoG)
- These model cells are arranged in specific patterns, mimicking the structure observed in the visual cortex
- The combination is done using a weighted geometric mean (detect edges and contours based on stimulus orientation.)



Mimic the simple cell behaviour

Via Push-Pull CORF operator

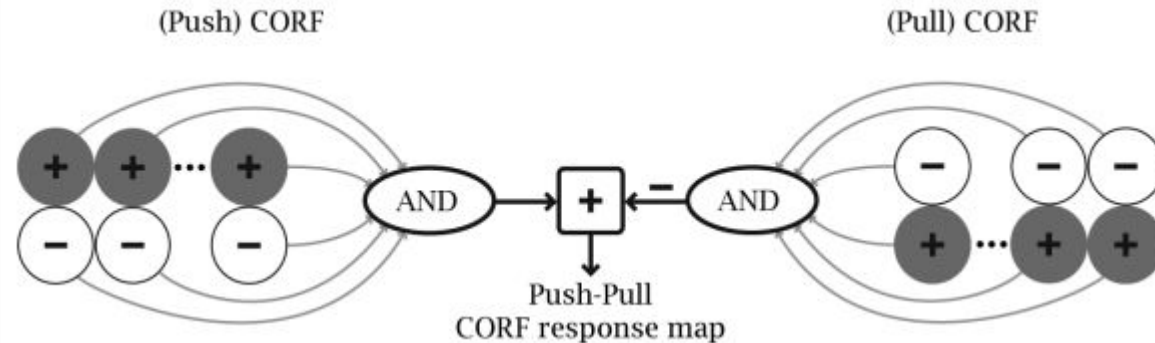
- Takes as input the response of two CORF models, their outputs are combined by a linear function.
"Push" response of a CORF model cell that is selective for a stimulus with preferred orientation and preferred contrast and "pull" a fraction of the response of a CORF model cell that responds to the same stimulus but of opposite contrast
- Final output highlights the contours in the image



Enhancing the simple cell behaviour

Via Push-Pull CORF operator

- More biologically plausible than traditional methods
- It outperforms many existing methods in contour detection tasks



Objective

Novel augmentation strategy

To enhance robustness
against
test-time/real-time
images



Objective

Novel augmentation strategy

To enhance robustness
against
test-time/real-time
images

Foster shape-biased learning

Via contour maps that
emphasize salient objects
using the push-pull CORF
operator

Objective

Novel augmentation strategy

To enhance robustness
against
test-time/real-time
images

Foster shape-biased learning

Via contour maps that
emphasize salient objects
using the push-pull CORF
operator

But how?

Create blended versions
of images with their
respective contour maps

Objective - Blend images with contour maps

But how?

Create blended versions
of images with their
respective contour maps

Objective - Blend images with contour maps



But how?

Create blended versions
of images with their
respective contour maps

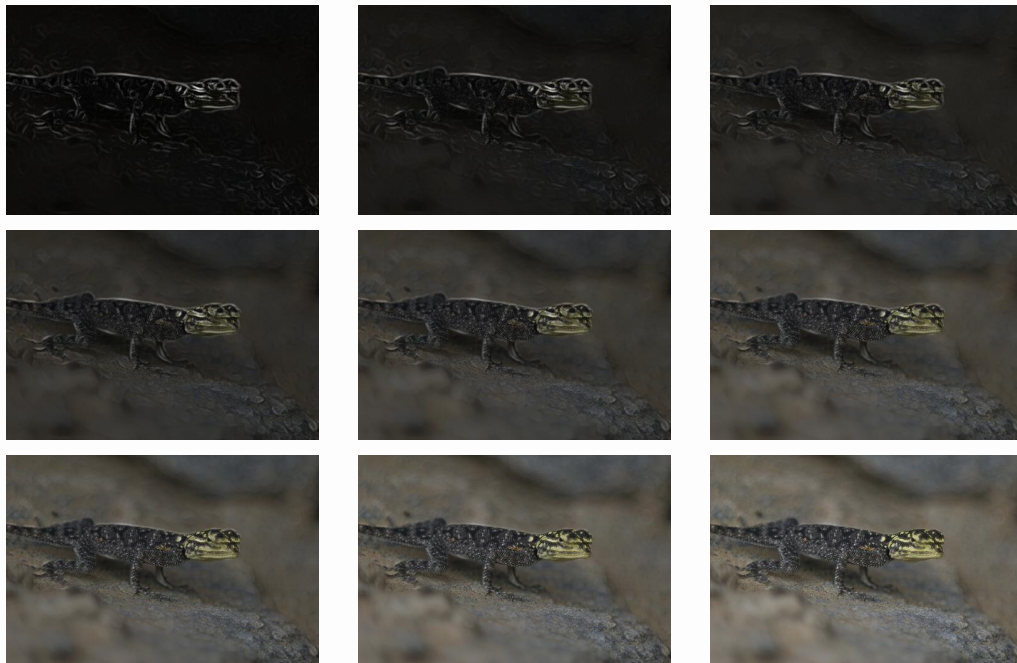
Objective - Blend images with contour maps



But how?

Create blended versions
of images with their
respective contour maps

Objective - Blend images with contour



Parameterized dataset blending

blend1, alpha = 0.1

.

.

blend9, alpha = 0.9

Parameterized dataset with blends

**Training on this
augmented
dataset –
BlendAug**

The model now trains on
the “clean images” and all
nine blended versions

Parameterized dataset with blends

Training on this augmented dataset – BlendAug

The model now trains on
the “clean images” and all
nine blended versions

Robust augmentation

But computationally and
time-expensive – doesn't
allow for further data
diversity

Parameterized dataset with blends

Training on this augmented dataset – BlendAug

The model now trains on
the “clean images” and all
nine blended versions

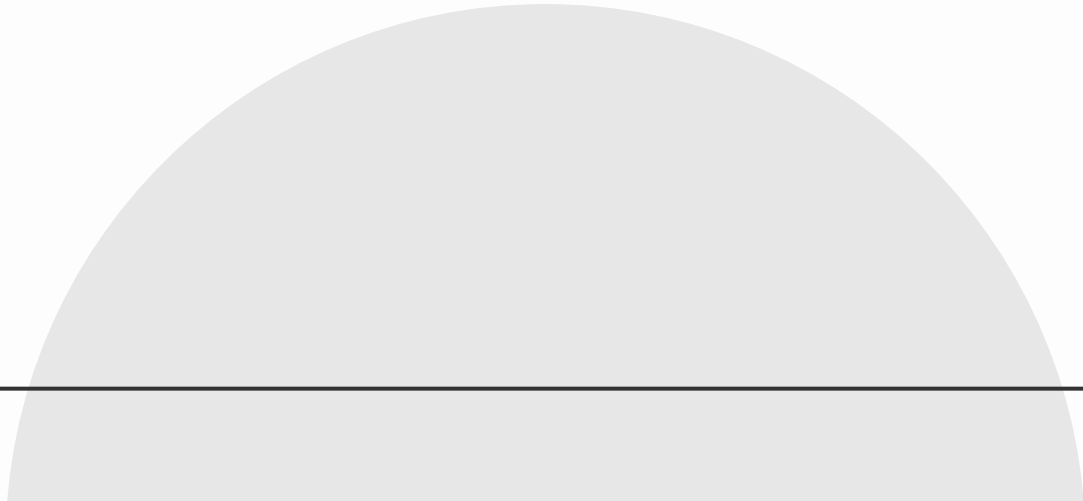
Robust augmentation

But computationally and
time-expensive – doesn’t
allow for further data
diversity

Enhancement in the training procedure

Ensure the training set is half
clean images/half random
selection from all blended
versions - OnlineAug

Methodology - BlendAug and OnlineAug



Methodology - BlendAug and OnlineAug



Strategy

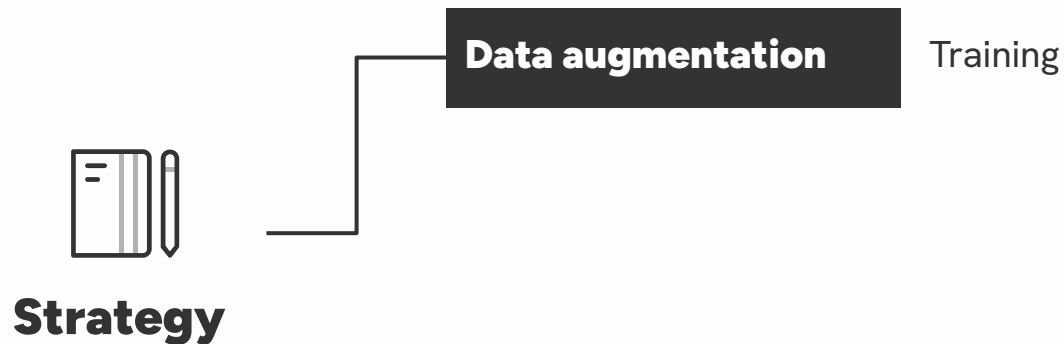
Methodology - BlendAug and OnlineAug



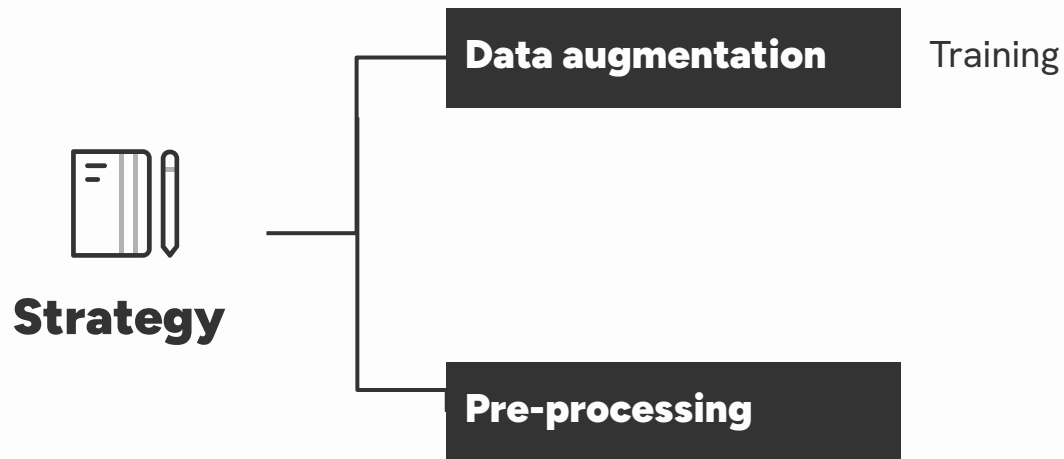
Strategy

Data augmentation

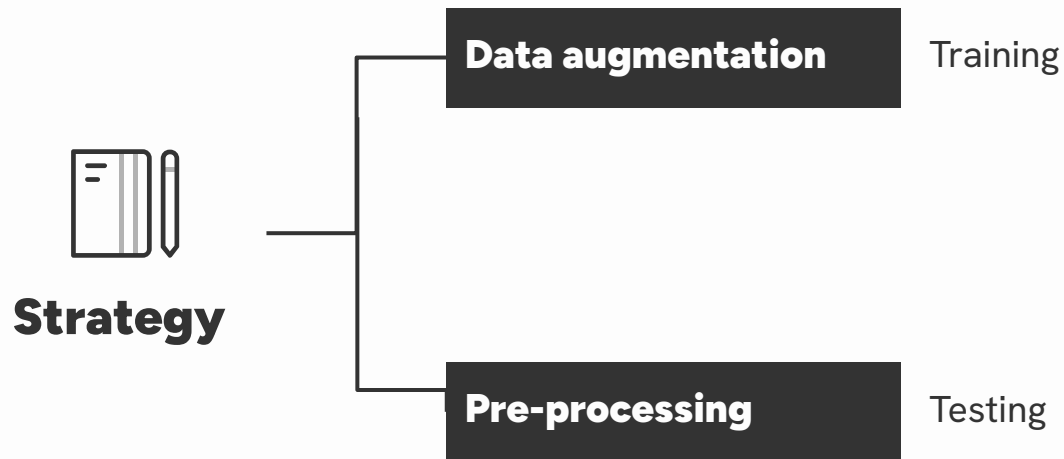
Methodology - BlendAug and OnlineAug



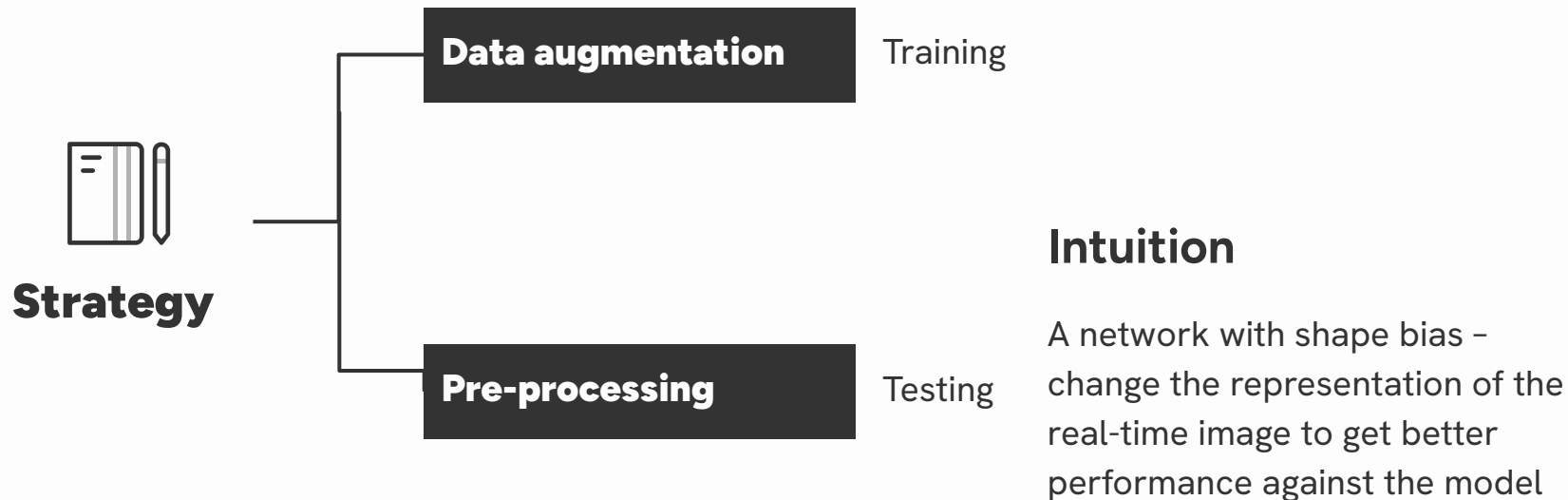
Methodology - BlendAug and OnlineAug



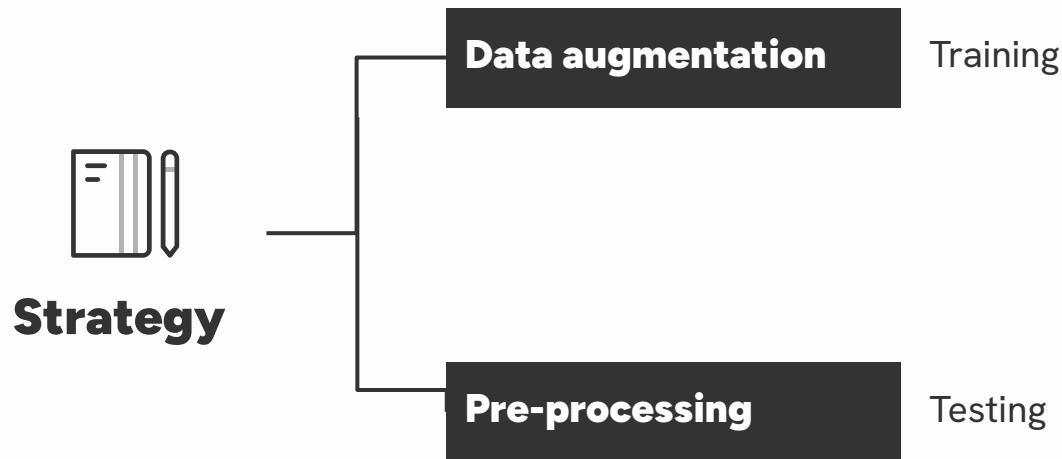
Methodology - BlendAug and OnlineAug



Methodology - BlendAug and OnlineAug



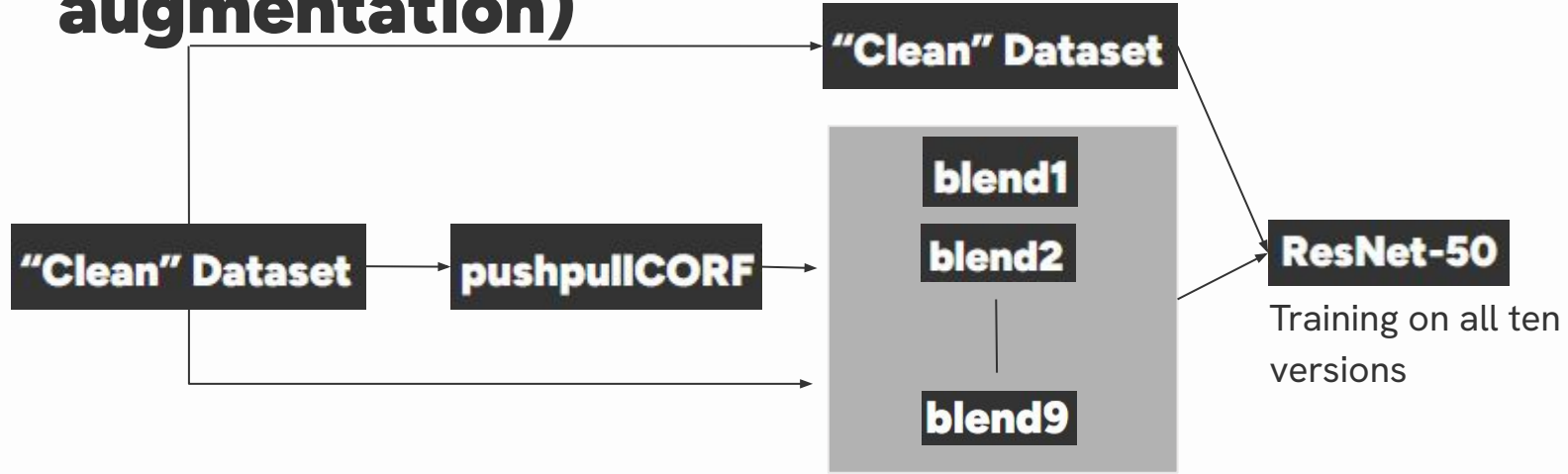
Methodology - BlendAug and OnlineAug



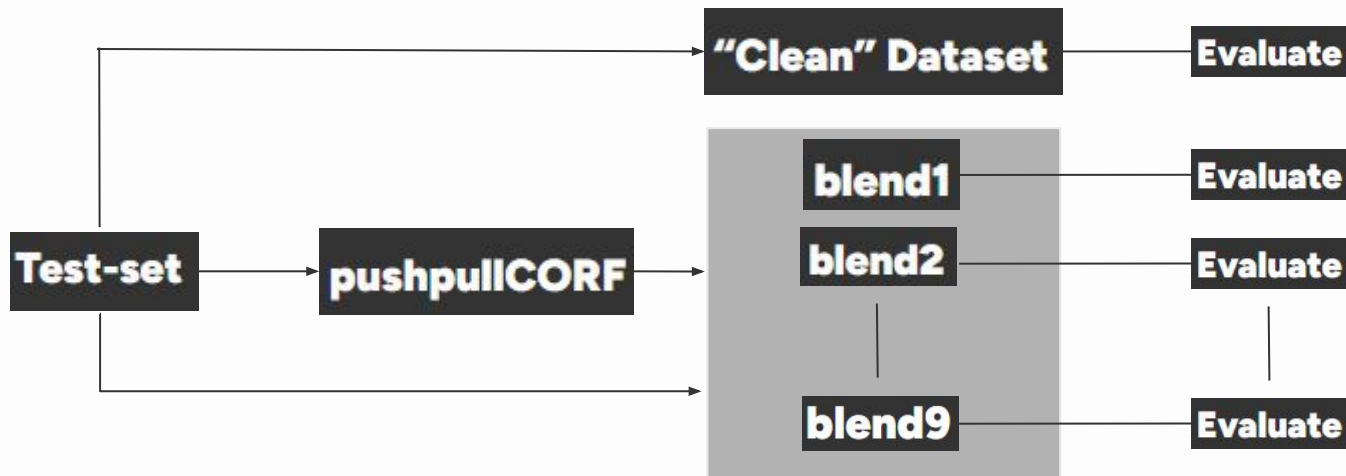
Intuition

A network with shape bias – change the representation of the real-time image to get better performance against the model i.e. create blended versions of real-time images and achieve better performance

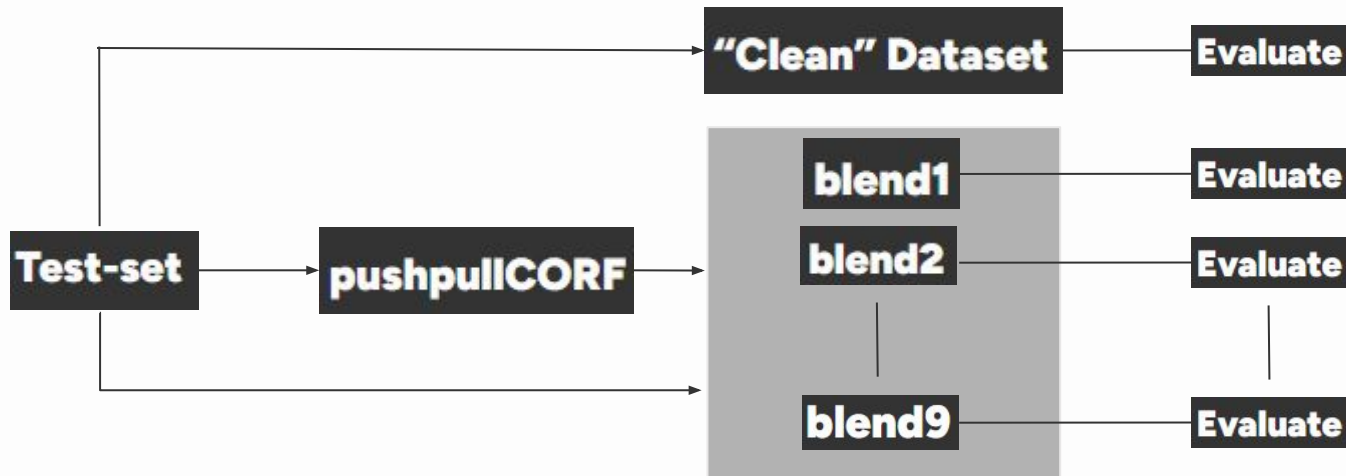
Methodology - BlendAug (data augmentation)



Methodology - BlendAug (pre-processing)

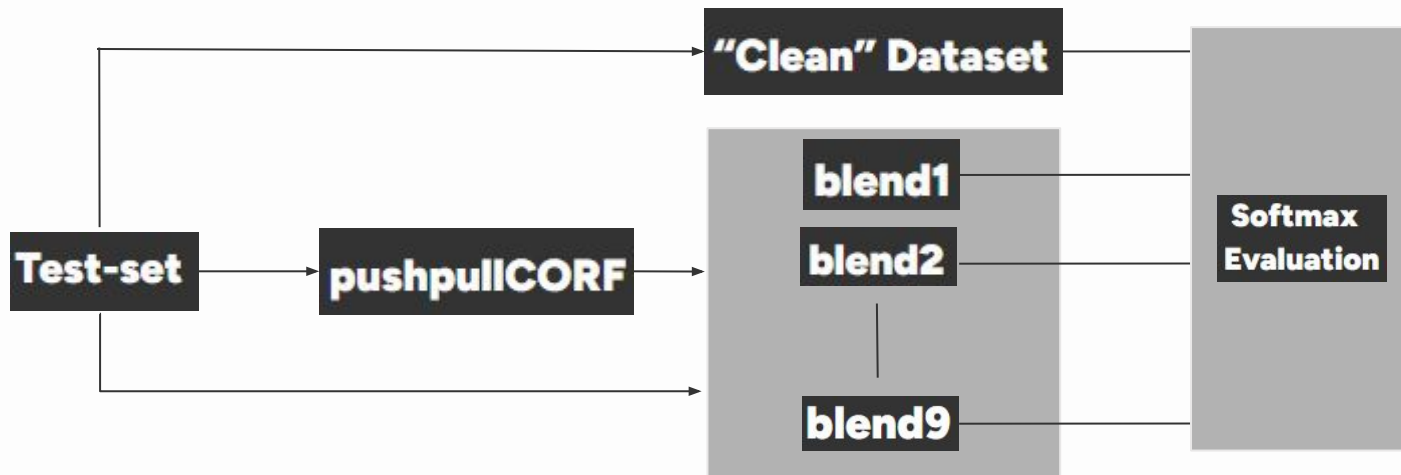


Methodology - BlendAug (pre-processing)

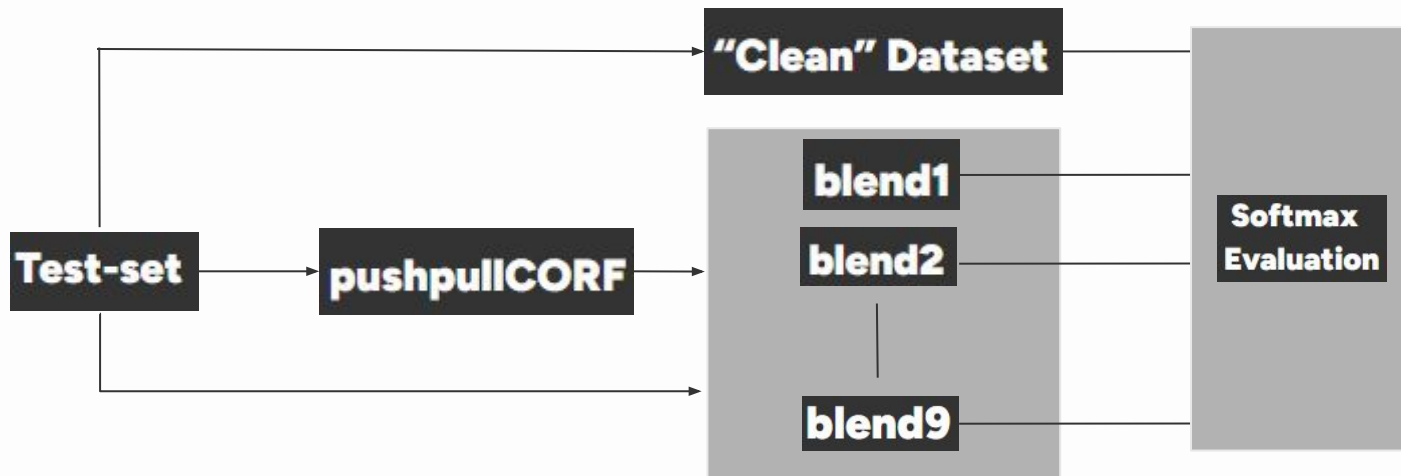


Test performance of the trained model against each blended versions - ten performance stats

Methodology - BlendAug (pre-processing)

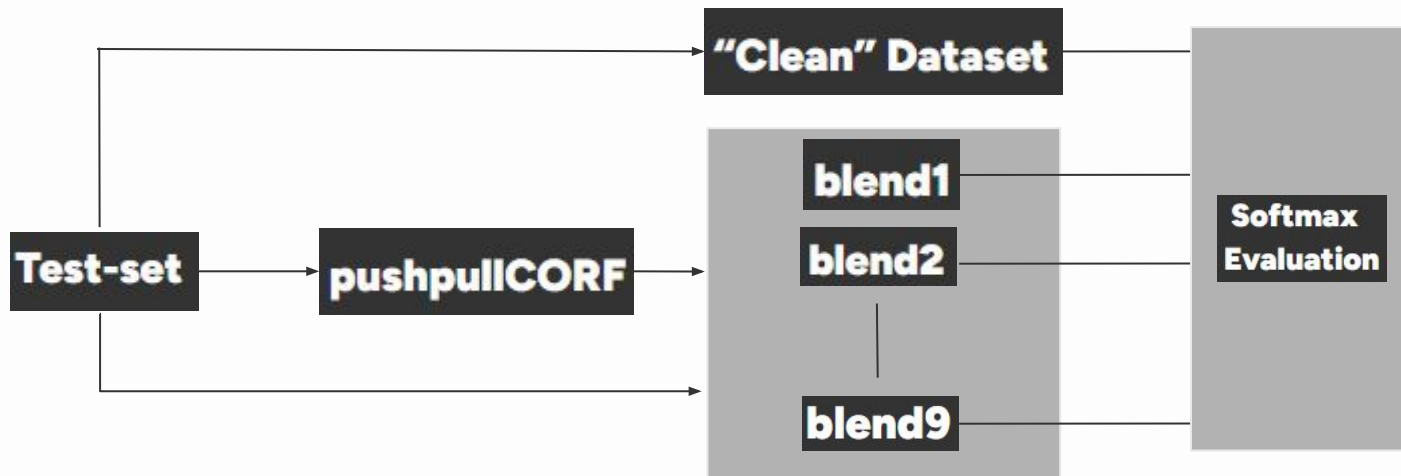


Methodology - BlendAug (pre-processing)



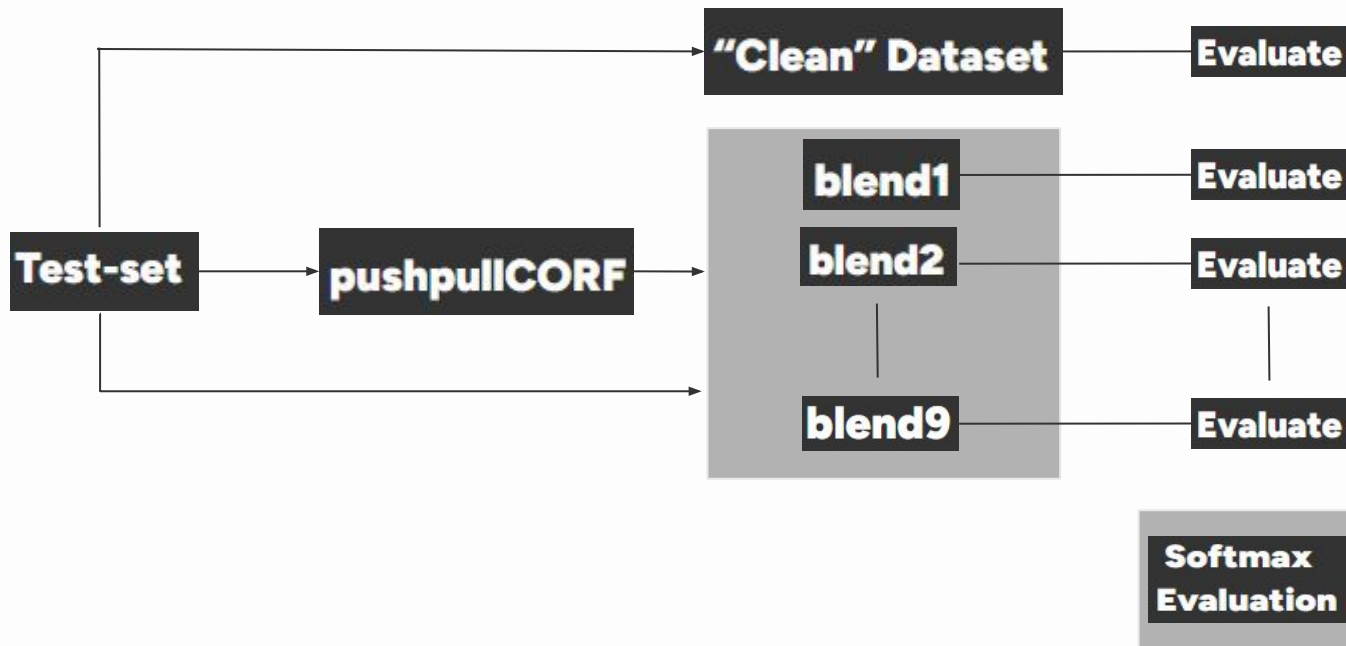
Classify one image, based on all ten representations

Methodology - BlendAug (pre-processing)



Calculate output against each image, convert logits to probabilities via softmax. Summate results (element-wise) and average by 10. Pick the class with highest probability

Methodology - BlendAug (pre-processing)



Total of 11 performance stats from BlendAug. Intuition is that the blended versions will enhance the performance of the trained model

Methodology - OnlineAug

Training on this augmented dataset – BlendAug

The model now trains on
the “clean images” and all
nine blended versions

Methodology - OnlineAug

Training on this augmented dataset – BlendAug

The model now trains on
the “clean images” and all
nine blended versions

Robust augmentation

But computationally and
time-expensive – doesn't
allow for further data
diversity

Methodology - OnlineAug

Training on this augmented dataset – BlendAug

The model now trains on the “clean images” and all nine blended versions

Robust augmentation

But computationally and time-expensive – doesn’t allow for further data diversity

Enhancement in the training procedure

Ensure the training set is half clean images/half random selection from all blended versions - OnlineAug

Methodology - OnlineAug

Introduce more blended versions

For further increasing the
data diversity

Methodology - OnlineAug

Introduce more blended versions

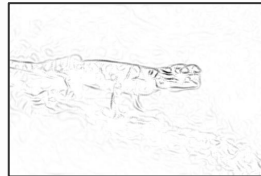
For further increasing the
data diversity



Methodology - OnlineAug

Introduce more blended versions

For further increasing the
data diversity



Methodology - OnlineAug

Introduce more blended versions

For further increasing the
data diversity - via
luminosity



Methodology - OnlineAug

Introduce more blended versions

Also parameterized -
alpha ranging from 0.1 to
0.9



Methodology - OnlineAug

**Introduce more
blended versions
– But this
increases
dataset size by
18x!**



Methodology - OnlineAug

**How to train
with such scale?**

Methodology - OnlineAug

How to train with such scale?

Include the clean dataset
as is



Methodology - OnlineAug

How to train with such scale?

Include the clean dataset
as is

Create a new dataset –
for every image in the
“clean” set, choose only
one blended version out
of the 18 available

Methodology - OnlineAug

How to train with such scale?

Include the clean dataset
as is

Create a new dataset –
for every image in the
“clean” set, choose only
one blended version out
of the 18 available

Effective dataset size
while training - **only 2x**
while providing all the
variations!

Methodology - OnlineAug

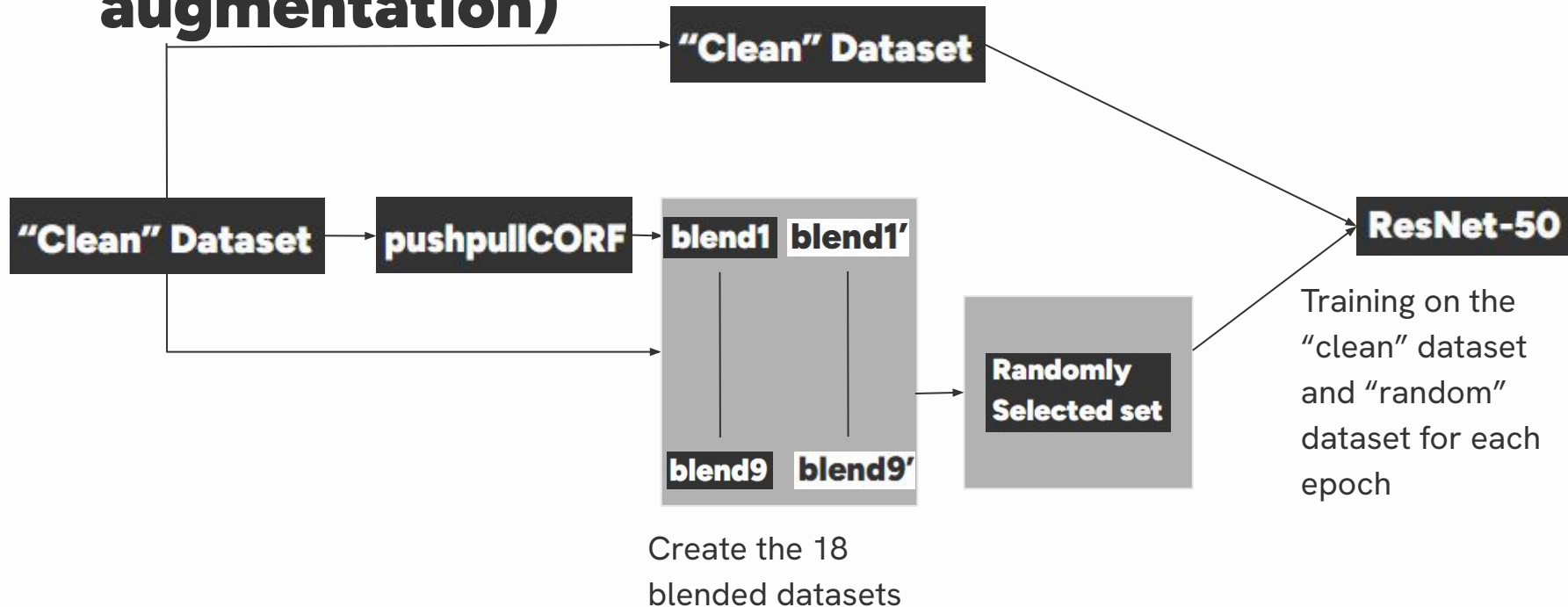
How to train with such scale?

Include the clean dataset
as is

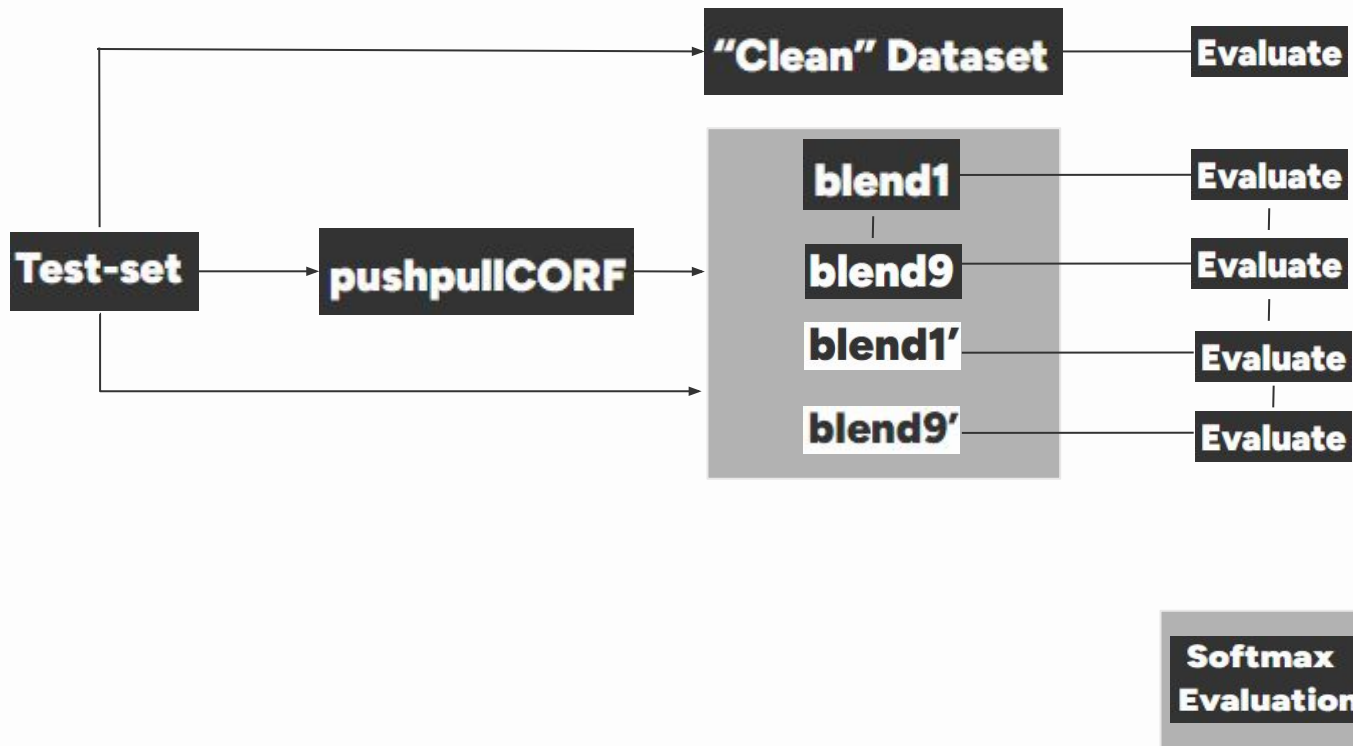
Create a new dataset –
for every image in the
“clean” set, choose only
one blended version out
of the 18 available

Effective dataset size
while training - **only 2x**
while providing all the
variations!
OnlineAug

Methodology - BlendAug (data augmentation)

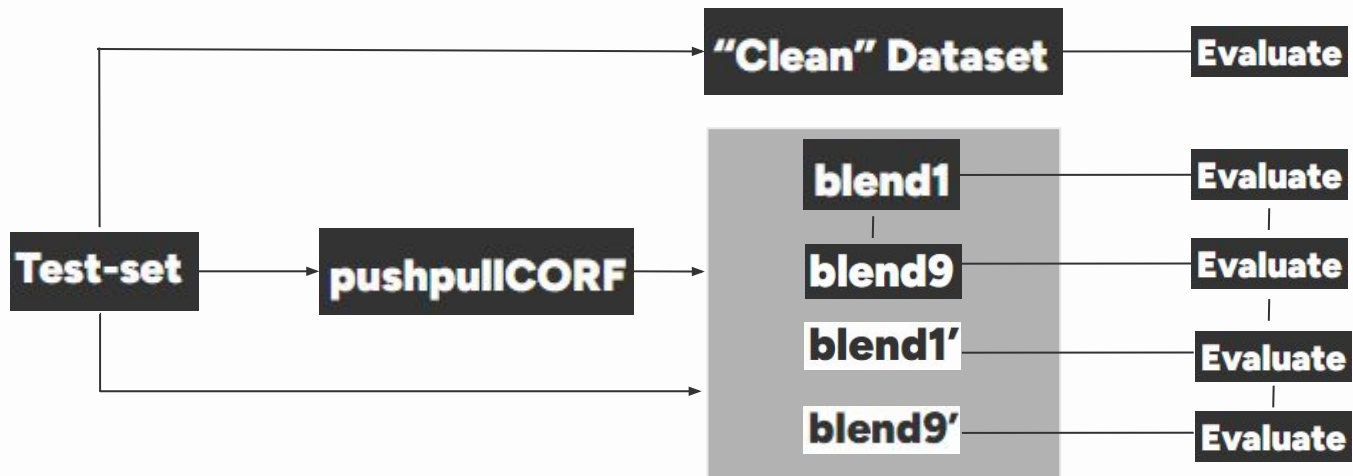


Methodology - OnlineAug (pre-processing)



Total of 19 performance stats from BlendAug. One softmax evaluation + evaluation on all 19 versions of the dataset

Methodology - OnlineAug (pre-processing)



**Softmax
Evaluation**

Softmax
evaluation takes
place against 19
versions of the
dataset

Experimental Setup

- BlendAug and OnlineAug

Experimental Setup - Dataset

Training Dataset

Experimental Setup - Dataset

Training Dataset

ImageNet-1k

Experimental Setup - Dataset

Training Dataset

ImageNet-1k - Train ResNet-50 (widely used for research)

Experimental Setup - Dataset

Training Dataset

ImageNet-1k - Train ResNet-50 (widely used for research)

Validation Dataset

Experimental Setup - Dataset

Training Dataset

ImageNet-1k - Train ResNet-50 (widely used for research)

Validation Dataset

Generalization capability analyzed via the following concerns

Experimental Setup - Dataset

Training Dataset

ImageNet-1k - Train ResNet-50 (widely used for research)

Validation Dataset

Generalization capability analyzed via the following concerns

ImageNet-A

Clean, unmodified images that
consistently fool classifiers -
adversality

Experimental Setup - Dataset

Training Dataset

ImageNet-1k - Train ResNet-50 (widely used for research)

Validation Dataset

Generalization capability analyzed via the following concerns

ImageNet-A

Clean, unmodified images that consistently fool classifiers - adversality

ImageNet-R

Non-photogenic representation of natural objects

Experimental Setup - Dataset

Training Dataset

ImageNet-1k - Train ResNet-50 (widely used for research)

Validation Dataset

Generalization capability analyzed via the following concerns

ImageNet-A

Clean, unmodified images that consistently fool classifiers - adversality

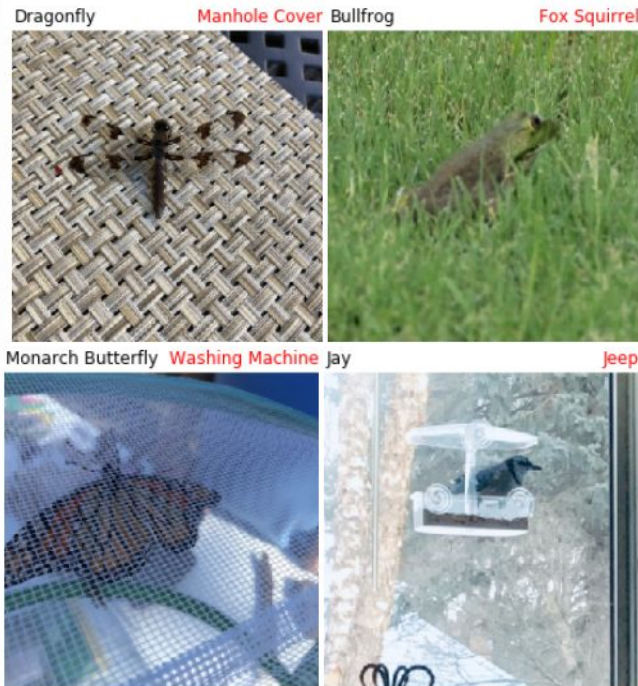
ImageNet-R

Non-photogenic representation of natural objects

ImageNet-C-bar

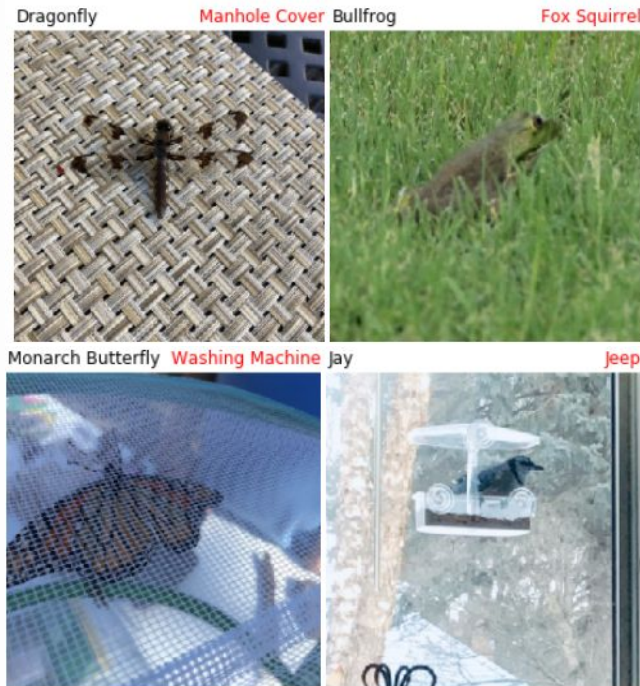
ImageNet-1k validation set exposed to ten different corruptions

Dataset - ImageNet-A



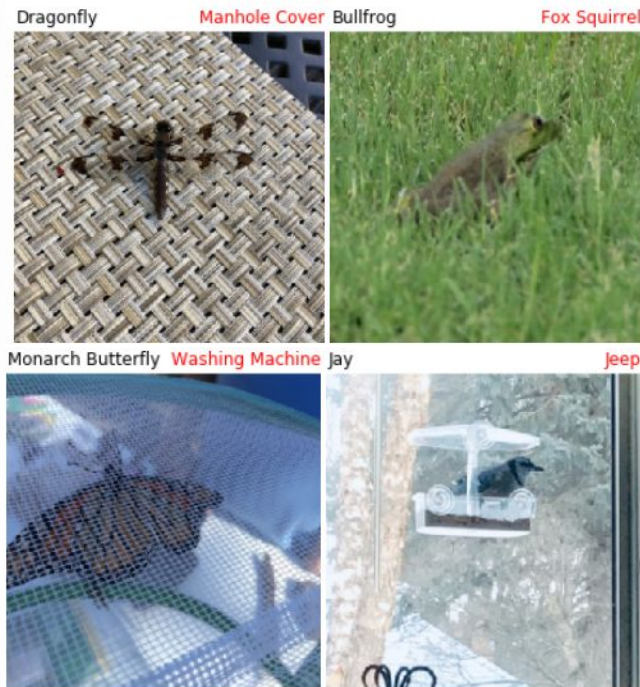
- Contains ~7,500 adversarially filtered images that consistently fool ImageNet classifiers

Dataset - ImageNet-A



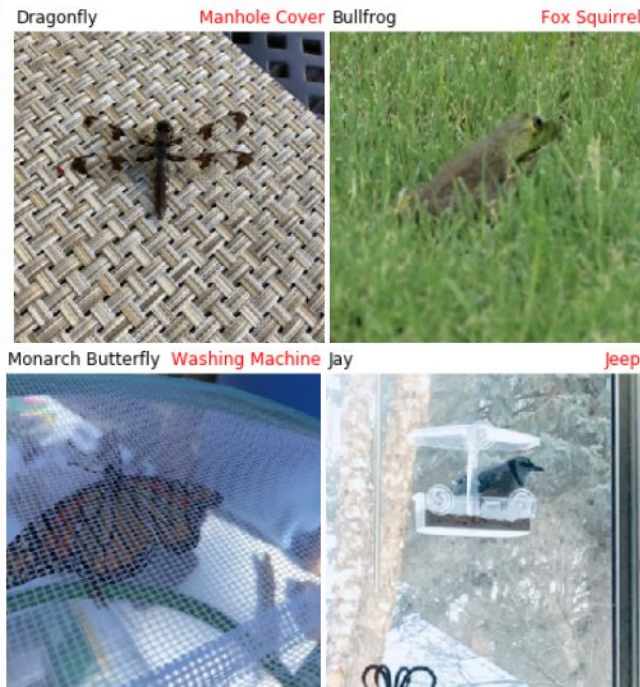
- Contains ~7,500 adversarially filtered images that consistently fool ImageNet classifiers
- Large number of images related to specific ImageNet classes from various online sources were collected for 200 broad classes

Dataset - ImageNet-A



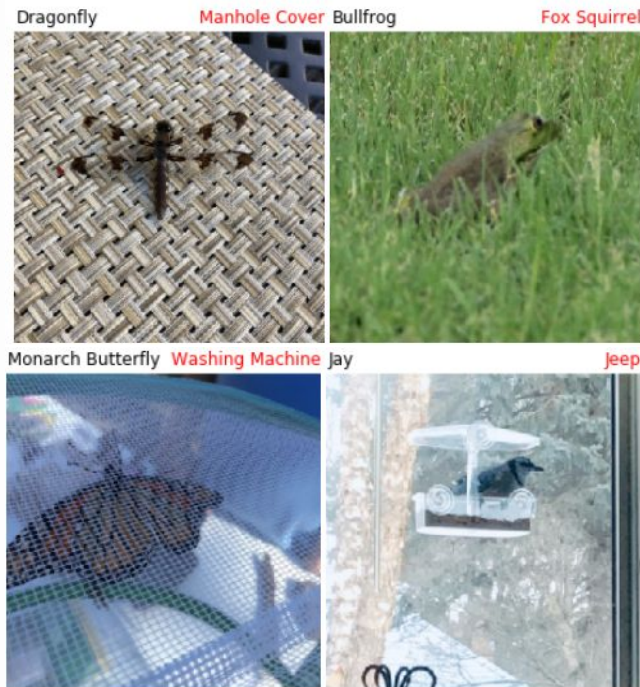
- Contains ~7,500 adversarially filtered images that consistently fool ImageNet classifiers
- Large number of images related to specific ImageNet classes from various online sources were collected for 200 broad classes
- A pre-trained ResNet-50 model used to classify – prune out correctly classified images

Dataset - ImageNet-A



- Contains ~7,500 adversarially filtered images that consistently fool ImageNet classifiers
- Large number of images related to specific ImageNet classes from various online sources were collected for 200 broad classes
- A pre-trained ResNet-50 model used to classify – prune out correctly classified images
- Authors manually selected visually clear and distinct images that belonged to a single class to ensure high-quality examples that could effectively challenge classifiers.

Dataset - ImageNet-A



- Contains ~7,500 adversarially filtered images that consistently fool ImageNet classifiers
- Large number of images related to specific ImageNet classes from various online sources were collected for 200 broad classes
- A pre-trained ResNet-50 model used to classify – prune out correctly classified images
- Authors manually selected visually clear and distinct images that belonged to a single class to ensure high-quality examples that could effectively challenge classifiers.
- Metric - Top-1 accuracy

Dataset - ImageNet-R



Painting

Sculpture



Origami

Cartoon



Embroidery

Toy

- Consists of 30,000 images representing renditions of 200 ImageNet classes.

Dataset - ImageNet-R



Painting

Sculpture



Origami

Cartoon



Embroidery

Toy

- Consists of 30,000 images representing renditions of 200 ImageNet classes.
- Compiled by sourcing diverse range of artistic renditions and non-photographic representations of images, including cartoons, graffiti, tattoos, sketches, etc

Dataset - ImageNet-R



Painting

Sculpture



Origami

Cartoon



Embroidery

Toy

- Consists of 30,000 images representing renditions of 200 ImageNet classes.
- Compiled by sourcing diverse range of artistic renditions and non-photographic representations of images, including cartoons, graffiti, tattoos, sketches, etc
- Collected images were filtered and labeled with the corresponding ImageNet class labels using MTurk annotators

Dataset - ImageNet-R



Painting

Sculpture



Origami

Cartoon



Embroidery

Toy

- Consists of 30,000 images representing renditions of 200 ImageNet classes.
- Compiled by sourcing diverse range of artistic renditions and non-photographic representations of images, including cartoons, graffiti, tattoos, sketches, etc
- Collected images were filtered and labeled with the corresponding ImageNet class labels using MTurk annotators
- Manually filtered the resulting images to ensure correct labeling and to verify that individual images did not contain multiple labels

Dataset - ImageNet-R



Painting

Sculpture



Origami

Cartoon



Embroidery

Toy

- Consists of 30,000 images representing renditions of 200 ImageNet classes.
- Compiled by sourcing diverse range of artistic renditions and non-photographic representations of images, including cartoons, graffiti, tattoos, sketches, etc
- Collected images were filtered and labeled with the corresponding ImageNet class labels using MTurk annotators
- Manually filtered the resulting images to ensure correct labeling and to verify that individual images did not contain multiple labels
- Dataset also incorporated line drawings from a previous study

Dataset - ImageNet-R



Painting

Sculpture



Origami

Cartoon

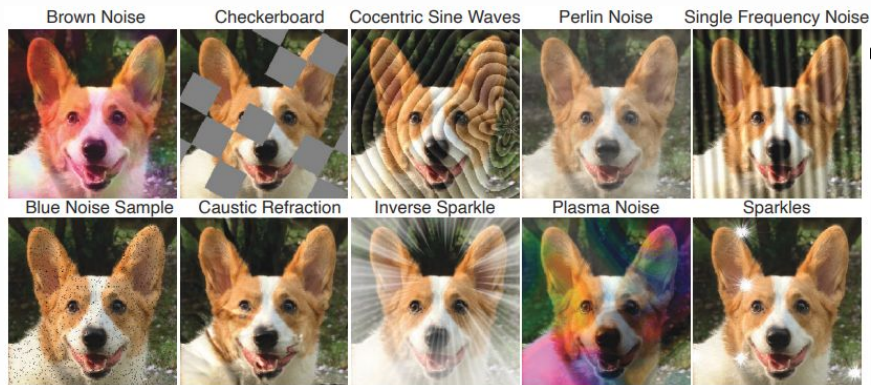


Embroidery

Toy

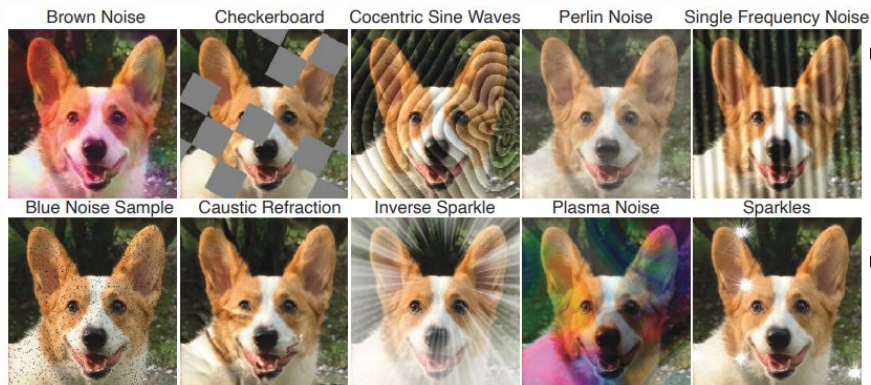
- Consists of 30,000 images representing renditions of 200 ImageNet classes.
- Compiled by sourcing diverse range of artistic renditions and non-photographic representations of images, including cartoons, graffiti, tattoos, sketches, etc
- Collected images were filtered and labeled with the corresponding ImageNet class labels using MTurk annotators
- Manually filtered the resulting images to ensure correct labeling and to verify that individual images did not contain multiple labels
- Dataset also incorporated line drawings from a previous study
- Metric Top-1 Error

Dataset - ImageNet-C-bar



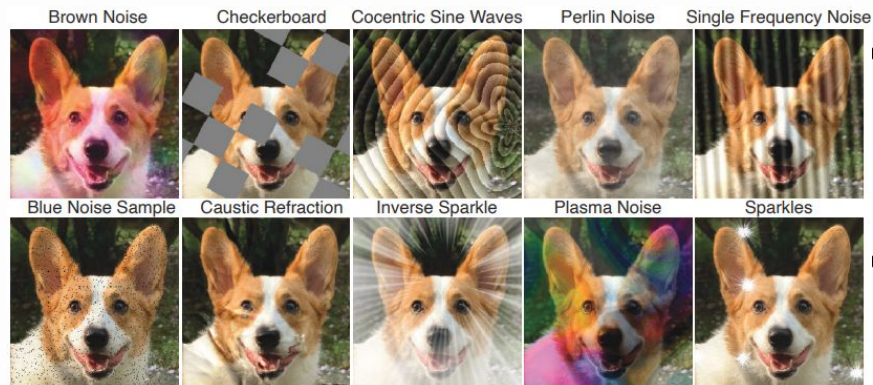
- Collection of image corruptions that are dissimilar from the standard ImageNet-C benchmark (corruptions like weather, blur, digital corruptions, etc)

Dataset - ImageNet-C-bar



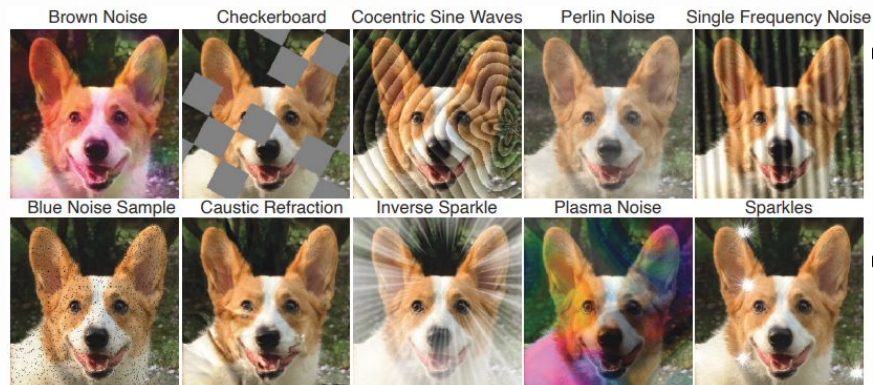
- Collection of image corruptions that are dissimilar from the standard ImageNet-C benchmark (corruptions like weather, blur, digital corruptions, etc)
- Consists of ten corruptions, (and similar to ImageNet-C) each corruption type implemented with five severity levels

Dataset - ImageNet-C-bar



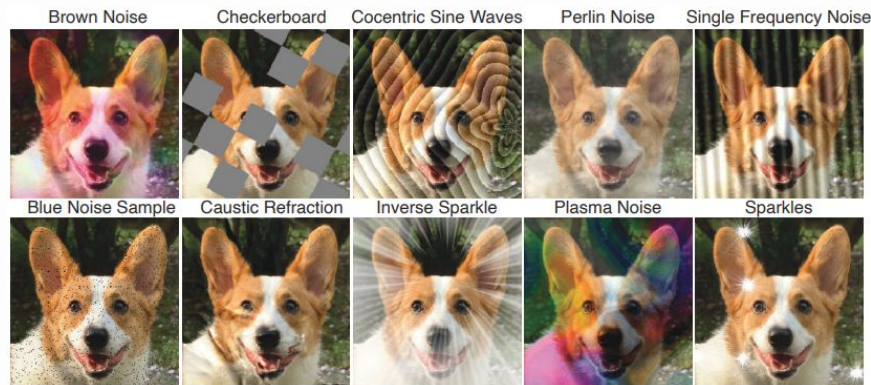
- Collection of image corruptions that are dissimilar from the standard ImageNet-C benchmark (corruptions like weather, blur, digital corruptions, etc)
- Consists of ten corruptions, (and similar to ImageNet-C) each corruption type implemented with five severity levels
- Metric

Dataset - ImageNet-C-bar



- Collection of image corruptions that are dissimilar from the standard ImageNet-C benchmark (corruptions like weather, blur, digital corruptions, etc)
- Consists of ten corruptions, (and similar to ImageNet-C) each corruption type implemented with five severity levels
- Metric
 - Corruption Error (CE) – for a severity, average of top-1 errors for each severity

Dataset - ImageNet-C-bar



- Collection of image corruptions that are dissimilar from the standard ImageNet-C benchmark (corruptions like weather, blur, digital corruptions, etc)
- Consists of ten corruptions, (and similar to ImageNet-C) each corruption type implemented with five severity levels
- Metric
 - Corruption Error (CE) – for a severity, average of top-1 errors for each severity
 - Mean Corruption Error (mCE) – average error rate across all corruption types and severity levels.

Experimental Setup - Training Recipe

- Following the standard training recipe of ResNet-50

Experimental Setup - Training Recipe

- Following the standard training recipe of ResNet-50 – isolate cause of improvement in performance solely for the employed strategy

Experimental Setup - Training Recipe

- Following the standard training recipe of ResNet-50 – isolate cause of improvement in performance solely for the employed strategy
- Recipe
 - 100 Epochs
 - CrossEntropyLoss
 - Batch size = 64
 - SGD with lr = 0.1, momentum = 0.9, weight decay = 0.0001
 - StepLR with step_size of 30 and gamma = 0.1

Experimental Setup - Training Recipe

- Following the standard training recipe of ResNet-50 - isolate cause of improvement in performance solely for the employed strategy
- Recipe
 - 100 Epochs
 - CrossEntropyLoss
 - Batch size = 64
 - SGD with lr = 0.1, momentum = 0.9, weight decay = 0.0001
 - StepLR with step_size of 30 and gamma = 0.1
- Retrieve model with the best performance against ImageNet-1k validation set after the training is complete

Results

ImageNet-A

ImageNet-A

Performance against
adversarial images

ImageNet-A

Performance against
adversarial images

Data Augmentation	Accuracy
Baseline	2.2
Adversarial Training	1.68
Stylized Transfer	2.33
OnlineAug + (softmax evaluation)	2.82
OnlineAug + (clean set evaluation)	2.97
OnlineAug + (blend6'/blend7'/blend8' evaluation)	2.99 (± 0.007)
AugMix	3.8
BlendAug + (softmax evaluation)	4.04
BlendAug + (clean set evaluation)	4.12
BlendAug + (blend8 evaluation)	4.20
Cutout	4.4
MoEx	5.5
Mixup	6.6
CutMix	7.3

ImageNet-A

Performance against
adversarial images

Data Augmentation	Accuracy
Baseline	2.2
Adversarial Training	1.68
Stylized Transfer	2.33
OnlineAug + (softmax evaluation)	2.82
OnlineAug + (clean set evaluation)	2.97
OnlineAug + (blend6'/blend7'/blend8' evaluation)	2.99 (± 0.007)
AugMix	3.8
BlendAug + (softmax evaluation)	4.04
BlendAug + (clean set evaluation)	4.12
BlendAug + (blend8 evaluation)	4.20
Cutout	4.4
MoEx	5.5
Mixup	6.6
CutMix	7.3

ImageNet-R

ImageNet-R

Performance against
non-photogenic renditions
of natural objects

ImageNet-R

Performance against
non-photogenic renditions
of natural objects

Data Augmentation	Error
Baseline	63.9
Adversarial Training	68.6
Speckle Noise	62.1
BlendAug + (clean set evaluation)	62.06
BlendAug + (blend5 evaluation)	61.42
OnlineAug + (clean set evaluation)	61.19
BlendAug + (softmax evaluation)	60.49
OnlineAug + (blend3 evaluation)	60.3
OnlineAug + (softmax evaluation)	58.98
AugMix	58.9
DeepAugment	57.8
DeepAugment + AugMix	53.2

ImageNet-R

Performance against
non-photogenic renditions
of natural objects

Data Augmentation	Error
Baseline	63.9
Adversarial Training	68.6
Speckle Noise	62.1
BlendAug + (clean set evaluation)	62.06
BlendAug + (blend5 evaluation)	61.42
OnlineAug + (clean set evaluation)	61.19
BlendAug + (softmax evaluation)	60.49
OnlineAug + (blend3 evaluation)	60.3
OnlineAug + (softmax evaluation)	58.98
AugMix	58.9
DeepAugment	57.8
DeepAugment + AugMix	53.2

ImageNet-C-bar

ImageNet-C-bar

Performance against ten
different corruptions

ImageNet-C-bar

Performance against ten
different corruptions

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checker board cutout	Cocn. Sin Waves	Inverse Sparkle s	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
Patch Gaussian	56.6	60.3	38.9	42.1	48.5	82.1	76.5	34.6	74.1	76.7	32.1
Stylized ImageNet	55.8	54.7	37.8	44.1	52.8	79.6	77.3	35.3	69.8	69.2	37
AutoAugment	55.7	54.8	34.5	43.8	43.8	86.5	77.1	33.8	68.3	78.8	36.1
Adversarial Noise Training	53.9	35.8	41	43.3	56.9	76.4	76.1	35.2	75.5	63.7	35
AugMix	52.4	43.2	38.8	39.2	46.1	76.3	76.4	32.4	72.2	67.4	32.3
DeepAugment	51	41.7	37.2	40.9	53.9	74.6	74.7	30.3	73.3	50.9	32.9
DeepAugment + AugMix	48.3	34.9	35.2	39.8	49.8	69.7	74.3	30.6	67.9	48	32.9
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

Performance against ten
different corruptions

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checker board cutout	Cocn. Sin Waves	Inverse Sparkle s	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
Patch Gaussian	56.6	60.3	38.9	42.1	48.5	82.1	76.5	34.6	74.1	76.7	32.1
Stylized ImageNet	55.8	54.7	37.8	44.1	52.8	79.6	77.3	35.3	69.8	69.2	37
AutoAugment	55.7	54.8	34.5	43.8	43.8	86.5	77.1	33.8	68.3	78.8	36.1
Adversarial Noise Training	53.9	35.8	41	43.3	56.9	76.4	76.1	35.2	75.5	63.7	35
AugMix	52.4	43.2	38.8	39.2	46.1	76.3	76.4	32.4	72.2	67.4	32.3
DeepAugment	51	41.7	37.2	40.9	53.9	74.6	74.7	30.3	73.3	50.9	32.9
DeepAugment + AugMix	48.3	34.9	35.2	39.8	49.8	69.7	74.3	30.6	67.9	48	32.9
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

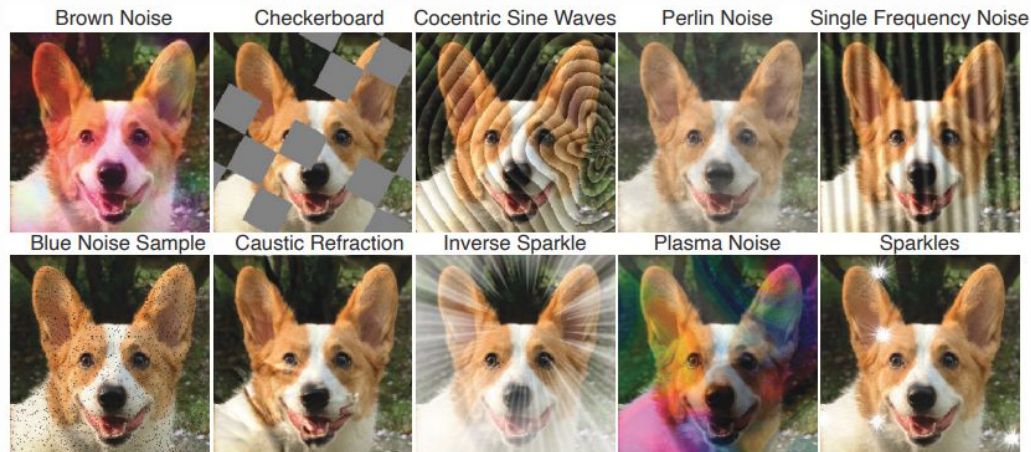
ImageNet-C-bar

Performance against ten
different corruptions

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refractio n	Checker board cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscal e	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

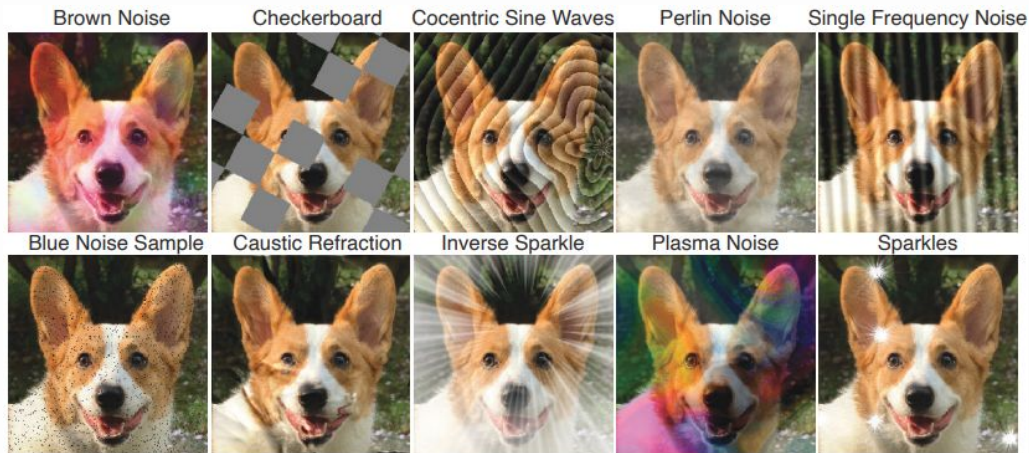
Performance against ten different corruptions



Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checkerboard cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

Performance against ten different corruptions

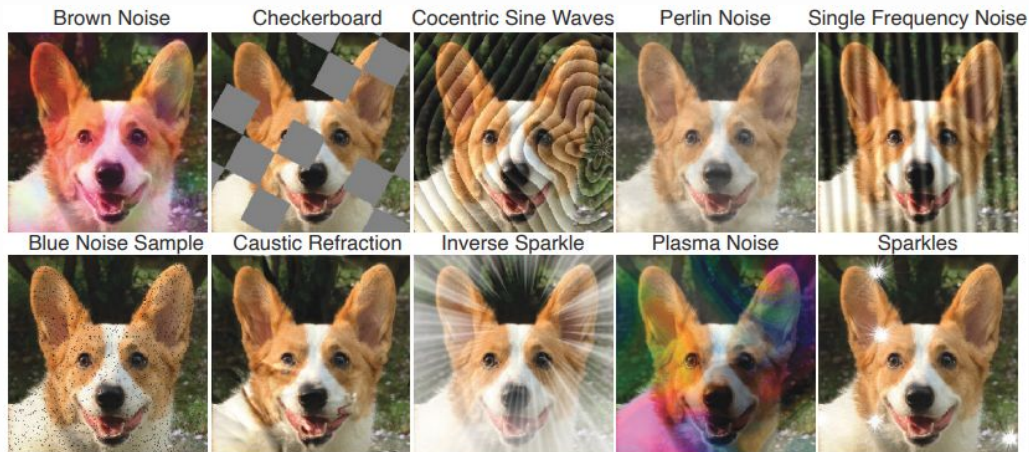


- High frequency noise
- Uniform ripple like distortions
- Single-frequency pattern, creating a uniform texture

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checkerboard cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

Performance against ten different corruptions



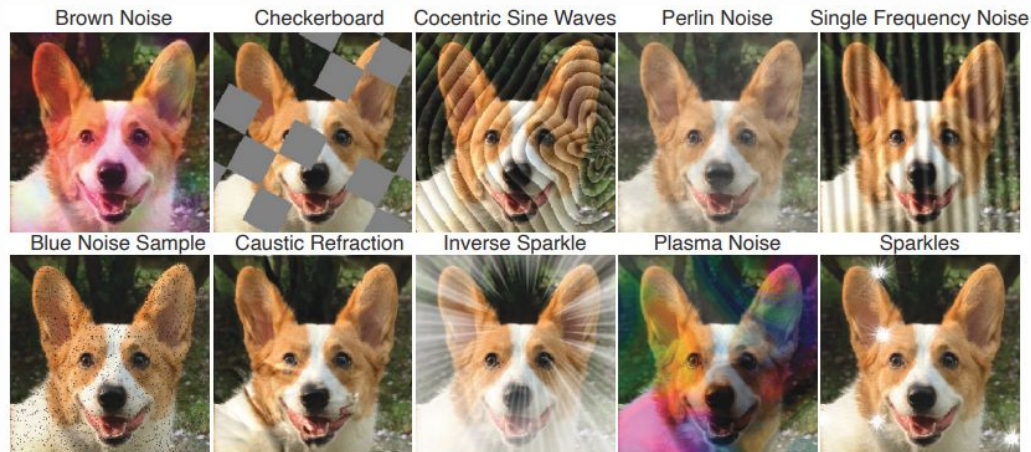
- High frequency noise
- Uniform ripple like distortions
- Single-frequency pattern, creating a uniform texture

=> better performance than baseline

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checkerboard cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

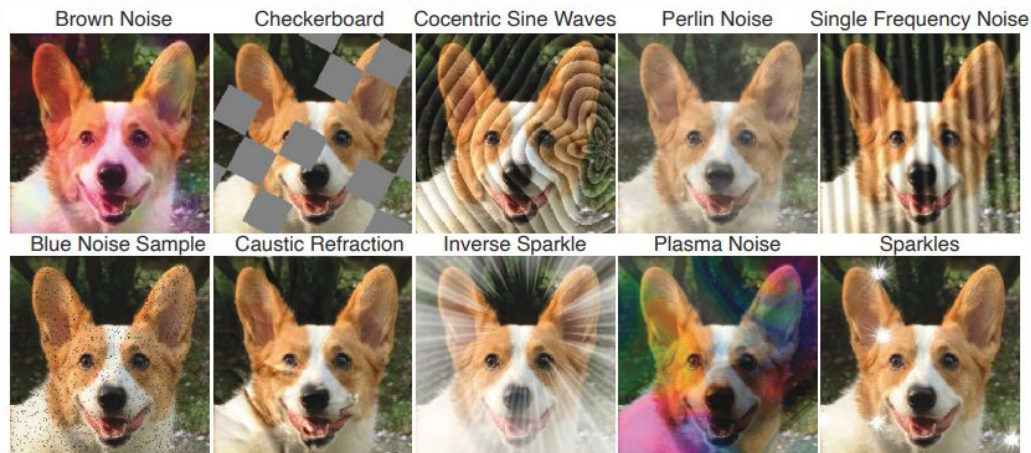
Performance against ten different corruptions



Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checkerboard cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

Performance against ten different corruptions

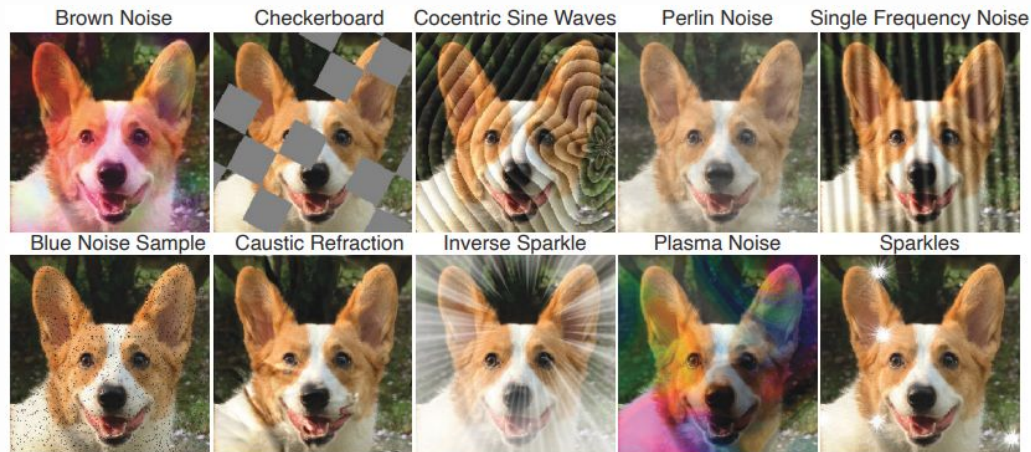


- Low frequency variations (br noise)
- Wavy-bright patterns (caustic refr)
- Removing parts of images ([inv] sparkles, checkr. cutout)
- Enhance texture (perlin noise)
- Color transitions (br + plasma noise)

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checkerboard cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

ImageNet-C-bar

Performance against ten different corruptions

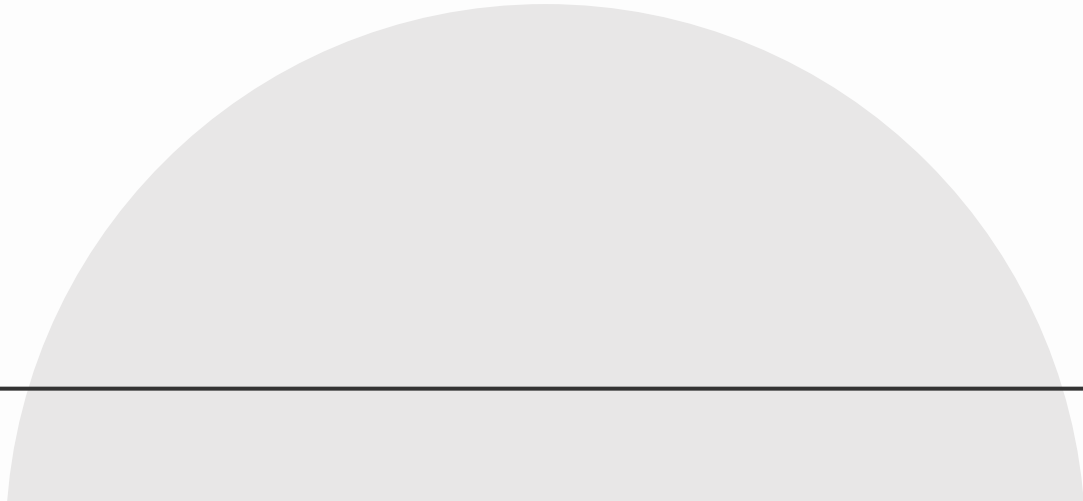


- Low frequency variations (br noise)
- Wavy-bright patterns (caustic refr)
- Removing parts of images ([inv] sparkles, checkr. cutout)
- Enhance texture (perlin noise)
- Color transitions (br + plasma noise)

=> worse performance than baseline

Data Augmentation	mCE	Blue Noise Sample	Brown. Noise	Caustic Refraction	Checkerboard cutout	Cocen. Sin Waves	Inverse Sparkles	Perlin Noise	Plasma Noise	Sin.Freq. Greyscale	Sparkles
Baseline	57.7	68.6	37.5	42.8	49.4	84.7	76.7	34.3	71.7	79	32.4
BlendAug + (clean set evaluation)	60.07	59.16	44.75	45.64	54.36	83.81	79.91	37.38	79.55	79.14	37.01
BlendAug + (softmax evaluation)	60.05	52.52	48.69	47.52	54.67	81.51	78.64	42.07	81.68	74.09	39.11
OnlineAug + (softmax evaluation)	59.86	53.6	48.4	47.87	53.96	81.15	79.2	40.66	81.05	73.56	39.15
OnlineAug + (clean set evaluation)	58.967	57.46	43.84	45.16	52.59	82.76	80.09	35.73	79.18	76.72	39.14

Conclusions and Future Work



Conclusions

**BlendAug and OnlineAug is a successful
strategy**

Conclusions

BlendAug and OnlineAug is a successful strategy

Improvement in performance when applied for augmentation and pre-processing purposes.

Conclusions

BlendAug and OnlineAug is a successful strategy

Improvement in performance when applied for augmentation and pre-processing purposes.

Achieving performance similar to state-of-the-art

Conclusions

BlendAug and OnlineAug is a successful strategy

Improvement in performance when applied for augmentation and pre-processing purposes.

Achieving performance similar to state-of-the-art

BlendAug and OnlineAug achieve performance similar to AugMix against adversarial and non-photogenic representations of ImageNet-like objects

Conclusions

BlendAug and OnlineAug is a successful strategy

Improvement in performance when applied for augmentation and pre-processing purposes.

Achieving performance similar to state-of-the-art

BlendAug and OnlineAug achieve performance similar to AugMix against adversarial and non-photogenic representative ImageNet objects.

Limited applicability against corruptions

Conclusions

BlendAug and OnlineAug is a successful strategy

Improvement in performance when applied for augmentation and pre-processing purposes

Achieving performance similar to state-of-the-art

BlendAug and OnlineAug achieve performance similar to AugMix against adversarial and non-photogenic representations. ImageNet results

Limited applicability against corruptions

The proposed methods don't bode well with various corruptions included in the study.

Future Work

Future Work

**Further understanding on the pre-processing
stages**

Future Work

Further understanding on the pre-processing stages

Specific version of the blended test set needed to achieve better performance, depending on the benchmark

Future Work

Further understanding on the pre-processing stages

Specific version of the blended test set needed to achieve better performance, depending on the benchmark

More insights on corruption robustness

Future Work

Further understanding on the pre-processing stages

Specific version of the blended test set needed to achieve better performance, depending on the benchmark

More insights on corruption robustness

Work against other benchmarks like ImageNet-C, to further understand the limitation of the strategy against corruption robustness, if present

Future Work

Further understanding on the pre-processing stages

Specific version of the blended test set needed to achieve better performance, depending on the benchmark

More insights on corruption robustness

Work against other benchmarks like ImageNet-C, to further understand the limitation of the strategy against corruption robustness, if present

Further analysis on generalization

Future Work

Further understanding on the pre-processing stages

Specific version of the blended test set needed to achieve better performance, depending on the benchmark

More insights on corruption robustness

Work against other benchmarks like ImageNet-C, to further understand the limitation of the strategy against corruption robustness, if present

Further analysis on generalization

Showing great potential against ImageNet-A and -R, evaluate against other notable benchmarks (like ImageNet-P, -O, etc) to test for wider applicability against real-time images

Thank you!

CREDITS: This presentation template was created by [Slidesgo](#)